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Oefeningenreeks 4A nr. 14.11

$$\begin{aligned} & \int \frac{\ln(\sin x)}{\cos^2 x} dx \\ & \left\{ \begin{array}{l} u = \ln(\sin x) \\ dv = \frac{1}{\cos^2 x} dx \end{array} \right. \Rightarrow \left\{ \begin{array}{l} du = \cot x \\ v = \tan x \end{array} \right. \\ & = \tan x \ln(\sin x) - \int \tan x \cdot \cot x dx \\ & = \tan x \ln(\sin x) - \int 1 dx \\ & = \tan x \ln(\sin x) - x + C \end{aligned}$$

References